Finite optical inverses and the error radius required for a shear constraint

Conditional methods review draft

A local distance law encodes expansion, acceleration and shear, but a catalogue measures that law through selected directions and nuisance projections. We compose these operations and derive the exact identifiable sector of the resulting finite linear response. Its inverse is a set: measurement and response errors determine a residual-compatible image, while the response kernel remains free. We give its support function, a stable inverse bound, and an explicit tensor ellipsoid with an additional bounded response term. For the complete positive angular magnitude field, normalized exponentiation instead gives an exact intercept-independent inverse; a uniform field error yields a finite bound on the recovered dipole and shear. These two inverses have different domains. A retained CF4 calibration quotient illustrates the finite construction with five dimensionless tensor coordinates, norm 0.0326568018932, and principal values approximately -0.0237272, 0.00132795 and 0.0223992. Conditional shear bounds become positive only below the apparent tensor norm. We show their dependence on an assumed response-error radius and Hubble normalization, retaining the tensor region and the physical sectors absent from the local slope.

I. WHAT THE LOCAL DISTANCE SLOPE CONTAINS

The first distance coefficient is a directional rate. Its physical content depends on the source congruence and observer, while its recovery depends on how the directions and distances were sampled. A finite tensor estimate therefore poses two questions: which combinations survive the observation operator, and which physical states remain compatible with its errors? We answer these questions with a composed response and a set-valued inverse.

General luminosity-distance cosmography expresses the local coefficients through finitely many kinematic multipoles [1]. Observer transformations must also be carried through the distance and redshift observables [2]. We use the same-congruence local law and logarithmic remainder established in the companion optical construction [3]. Our finite result retains the sampling and nuisance operators before attempting an inverse. Its error regions are deterministic sets unless a joint sampling law is explicitly supplied.

Use signature $(-+++)$, proper length $s = c\tau$, and a unit timelike congruence $u^a u_a = -1$. Let n point from observer to source, with past-directed null direction $\ell = -u + n$. In the orthonormal rest space at the observation event,

$$\nabla_a u_b = -u_a A_b/c^2 + (H h_{ab} + \sigma_{ab} + \omega_{ab})/c, \qquad h_{ab} = g_{ab} + u_a u_b. \tag{1}$$

Here H, σ, ω have units of inverse time and A of length per time squared. Since $\ell^a \ell^b \omega_{ab} = 0$,

$$\mathcal{H}_u(n) = c\ell^a \ell^b \nabla_a u_b = H - A \cdot n/c + \sigma : nn, \qquad d_L(z, n) = \frac{cz}{\mathcal{H}_u(n)} + O(z^2). \tag{2}$$

Equation (2) assumes smooth emitters and an observer following this congruence, with a regular local geometric-optics distance expansion. The acceleration sign follows from our direction convention. Finite-distance peculiar motion, frame changes and calibration enter the application response separately.

For $H > 0$, define the dimensionless vector and STF tensor

$$d = -A/(cH), \quad S = \sigma/H, \quad q_\theta(n) = d \cdot n + S : nn, \quad \theta = (d, S), \quad \kappa = 5/\ln 10. \tag{3}$$

The local angular magnitude field is

$$f_\theta(n) = \lim_{z \downarrow 0} 5 \log_{10} \frac{d_L(z, n)}{cz/H} = -\kappa \log[1 + q_\theta(n)], \qquad \inf_n [1 + q_\theta(n)] > 0. \tag{4}$$

The ratio has a finite limit even though each magnitude diverges. The absolute scale H is absent from (4). Multiplying (H, A, σ) by the same positive constant preserves (d, S) and this angular field.

II. AN INVERSE FOR THE COMPLETE ANGULAR FIELD

Let $\langle \cdot \rangle$ denote normalized full-sphere integration. The following inverse requires the complete field and this measure.

Proposition 1 (intercept-independent inverse and a finite field-error bound). For any constant intercept b, set

$$r_f(n) = \frac{e^{-(f_\theta(n) + b)/\kappa}}{\langle e^{-(f_\theta + b)/\kappa} \rangle}. \tag{5}$$

Then $r_f = 1 + q_\theta$, and

$$d_i = 3\langle r_f n_i\rangle, \qquad S_{ij} = \frac{15}{2}\langle r_f(n_i n_j - \delta_{ij}/3)\rangle. \tag{6}$$

Suppose a complete perturbed field is $f_\theta + b + u$, with $\|u\|_\infty \le \delta$, and $\|q_\theta\|_\infty \le \epsilon < 1$. Apply (5)–(6) to that field to obtain $(\widetilde{d}, \widetilde{S})$. Then

$$|\widetilde{d} - d| \le \sqrt{3}B, \qquad \|\widetilde{S} - S\|_F \le \sqrt{15/2}B, \qquad B = (1+\epsilon)(e^{2\delta/\kappa} - 1). \tag{7}$$

Proof. Trace freedom and spherical symmetry give $\langle q_\theta\rangle = 0$, so the denominator in (5) is exactly $e^{-b/\kappa}$. The identities $\langle n_i n_j\rangle = \delta_{ij}/3$ and $\langle n_i n_j n_k n_l\rangle = (\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})/15$ prove (6). Write $r = 1 + q_\theta$. For the perturbed field,

$$\frac{\widetilde{r}}{r} = \frac{e^{-u/\kappa}}{\langle r e^{-u/\kappa}\rangle} \in [e^{-2\delta/\kappa}, e^{2\delta/\kappa}]. \tag{8}$$

The denominator is an average with positive normalized weight r, which proves this interval. Hence $\|\widetilde{r} - r\|_2 \le B$ in normalized spherical L^2. Orthogonal harmonic projection cannot increase that norm. A dipole has squared spherical norm $|d|^2/3$, and an STF quadrupole has squared norm $2\|S\|_F^2/15$. Applying these identities to the two projected differences proves (7). A constant u cancels exactly, although the uniform bound does not exploit that special case. □

This is also an injectivity result: two complete fields differing only by an intercept have the same d, S. Replacing the field by a truncated fit introduces the omitted field as an additional u; replacing the sphere by selected directions changes the integration operator. Without a bound on those changes, exponentiating a noisy low-order map does not inherit Proposition 1.

III. SAMPLING, NUISANCE REMOVAL AND THE OPTICAL RESPONSE

Let $\mathcal{S}$ be a specified finite sampling operator from angular fields to m magnitude coordinates. Point evaluations and normalized, possibly signed, window averages are examples. We require each row to have finite total variation ℓ_i, so $|(\mathcal{S}g)_i| \le \ell_i \|g\|_\infty$. Let $N\beta$ be a finite linear nuisance family in those same coordinates and let W be a nonsingular working whitening matrix. Using W as a metric imposes no stochastic law. Define

$$U = WN, \qquad P_N = I - UU^+, \qquad z = P_N W y, \qquad \Phi(\theta) = P_N W \mathcal{S} f_\theta. \tag{9}$$

The superscript + denotes the Moore–Penrose inverse. Since $P_N W N = 0$, the observation equation becomes

$$y = \mathcal{S} f_\theta + N\beta + e_0, \qquad z = \Phi(\theta) + P_N W e_0. \tag{10}$$

This is the composed nonlinear map. Its exact equivalence relation is $\theta \sim \theta'$ when $P_N W \mathcal{S}(f_\theta - f_{\theta'}) = 0$. Global injectivity of (5) does not establish injectivity of (9).

To expose a constructive finite inverse, choose Frobenius-orthonormal coordinates for S, and write $V\theta = q_\theta$. In the domain $\|q_\theta\|_\infty \le \epsilon < 1$,

$$f_\theta = -\kappa V\theta + R_1(\theta), \qquad \|R_1(\theta)\|_\infty \le b_1(\epsilon) := \frac{\kappa\epsilon^2}{2(1-\epsilon)}. \tag{11}$$

Indeed, the tail of $\log(1+q)$ after its linear term has absolute value at most $\sum_{k\ge 2} \epsilon^k/k \le \epsilon^2/[2(1-\epsilon)]$. Set

$$A = -\kappa P_N W \mathcal{S} V, \qquad z = A\theta + e, \quad e = P_N W[e_0 + \mathcal{S} R_1(\theta)]. \tag{12}$$

If $e_0 \in \mathcal{E}_0$, a uniform finite outer error region is

$$\mathcal{E} = P_N W \mathcal{E}_0 \;\oplus\; P_N W \prod_{i=1}^{m} [-\ell_i b_1, \ell_i b_1]. \tag{13}$$

For a compact set E, write $h_E(a) = \sup_{e\in E} a^{\mathsf{T}} e$. Equation (13) has the computable support

$$h_{\mathcal{E}}(a) = h_{\mathcal{E}_0}(W^{\mathsf{T}} P_N a) + b_1 \sum_i \ell_i |(W^{\mathsf{T}} P_N a)_i|. \tag{14}$$

The optical tail is correlated across directions and depends on θ. The box in (13) deliberately enlarges its image; (14) is exact for that outer set. The exact nonlinear feasible set instead retains the dependence,

$$\Theta_{\mathrm{nl}}(z) = \{\theta : \inf_n (1 + q_\theta) > 0, \quad z - \Phi(\theta) \in P_N W \mathcal{E}_0\}. \tag{15}$$

On the stated ϵ domain, (12)–(13) contain (15). A fit of only five shear columns is obtained by placing the chosen dipole and intercept columns in N. Their linear contribution is then removed, while their nonlinear coupling remains in R_1. This separation prevents nuisance projection from being mistaken for removal of every physical effect of those fields.

IV. THE EXACT FINITE IDENTIFIABLE SECTOR

We now analyze the declared linear error model $z = A\theta + e$, $e \in \mathcal{E}$, without imposing an additional physical restriction on θ. Let

$$A = U_r D_r V_r^{\mathsf{T}}, \qquad P_A = AA^{+}, \qquad P_I = A^{+}A = V_r V_r^{\mathsf{T}}, \qquad \mathcal{E}_z = \{e \in \mathcal{E} : (I - P_A)(z - e) = 0\}. \tag{16}$$

Only positive singular values occur in D_r. A numerical rank policy is a separately recorded approximation to this exact algebraic definition.

Theorem 2 (finite inverse and support). The model is feasible exactly when $\mathcal{E}_z \neq \varnothing$. Its full parameter set is

$$\Theta_{\mathrm{lin}}(z) = \{A^{+}(z - e) + k : e \in \mathcal{E}_z,\ k \in \ker A\}. \tag{17}$$

A linear carrier $J\theta$ is determined by the noiseless observation for all θ precisely when $\ker A \subseteq \ker J$. For such a carrier, its error image is compact when $\mathcal{E}$ is compact, and

$$h_{J\Theta_{\mathrm{lin}}}(a) = a^{\mathsf{T}} J A^{+} z + h_{\mathcal{E}_z}(-A^{+\mathsf{T}} J^{\mathsf{T}} a). \tag{18}$$

We use the extended-real convention $h_{\varnothing}(a) = -\infty$; thus an empty feasible set has an empty carrier image with support $-\infty$. If $\mathcal{E}_z \neq \varnothing$ and $J^{\mathsf{T}} a$ has a nonzero component in $\ker A$, the support in that direction is $+\infty$. For $r > 0$, if $\|e\|_2 \leq r_e$, the identifiable coordinate obeys

$$\|P_I \theta - A^{+} z\|_2 \leq \frac{r_e}{s_{\min}(A)}. \tag{19}$$

Proof. If $z - e = A\theta$, multiplication by $I - P_A$ gives the compatibility condition in (16), and multiplication by A^{+} gives $P_I\theta = A^{+}(z - e)$. The remaining component is in $\ker A$, proving necessity in (17). Conversely, every vector in (17) satisfies $A\theta = P_A(z - e) = z - e$. In the noiseless model two parameters have the same image exactly when their difference is in $\ker A$; applying J proves the stated criterion. Under that criterion, the kernel term vanishes and maximizing the affine functional of e gives (18). Compactness follows because $\mathcal{E}_z$ is a closed subset of a compact set. When $\mathcal{E}_z \neq \varnothing$, choose a feasible $\theta_0 \in \Theta_{\mathrm{lin}}(z)$. A nonzero kernel component provides $k \in \ker A$ with $a^{\mathsf{T}} J k > 0$;

then $\theta_0 + tk$ remains feasible for every $t \geq 0$, and $a^{\mathsf{T}} J(\theta_0 + tk) \to +\infty$, proving the infinite-support statement. Finally $P_I\theta - A^+ z = -A^+ e$, and $\|A^+\|_2 = 1/s_{\min}(A)$, which proves (19). □

For $A = 0$, every parameter is unresolved whenever $z \in \mathcal{E}$; (19) is undefined. If a physical domain Θ_0 is added, its feasible set is $\Theta_{\mathrm{lin}} \cap \Theta_0$. Domain restrictions can bound kernel directions, but that information comes from the domain. The exact support is then that of the intersection, with (18) providing an outer bound. Likewise, the rank of the tangent $A = D\Phi(0)$ does not assert global identifiability of (15).

Residual compatibility matters. Sending every $e \in \mathcal{E}$ through A^+ while ignoring $(I - P_A)(z - e) = 0$ can report a nonempty inverse for an inconsistent observation. This is the finite counterpart of carrying the unobserved angular field in Proposition 1.

A. An explicit tensor image

For a fully identified five-coordinate STF estimate, suppose a chosen coefficient-error model is

$$\widehat{b} = b + C^{1/2}u + Gv, \qquad \|u\|_2 \leq \rho, \quad \|v\|_2 \leq \eta, \qquad C \succeq 0. \tag{20}$$

Here C is an adopted dimensionless shape matrix, G maps the response-error coordinates to coefficient space, and both radii are specified externally. Under this independent product of error balls, the exact coefficient region and support are

$$\mathcal{B} = \widehat{b} + C^{1/2}B_\rho \oplus GB_\eta, \qquad h_{\mathcal{B}}(a) = a^{\mathsf{T}}\widehat{b} + \rho\sqrt{a^{\mathsf{T}}Ca} + \eta\|G^{\mathsf{T}}a\|_2. \tag{21}$$

The error sets are centrally symmetric, explaining the plus sign. The support follows by maximizing independently over u, v; Cauchy–Schwarz attains each maximum, including the zero-response cases by continuity. An arbitrary joint error region need not have this product structure. With an orthonormal STF basis T_α, the tensor is $S = \sum_\alpha b_\alpha T_\alpha$, and a tensor functional $L : S$ uses $a_\alpha = L : T_\alpha$ in (21). The full tensor region is preserved by this isometry.

A sufficient Frobenius radius containing (21) is

$$r_Q = \rho\sqrt{\lambda_{\max}(C)} + \eta\|G\|_2. \tag{22}$$

It discards anisotropic shape and can be conservative. If a fixed Gaussian joint coefficient law is additionally justified, one may select a quantile for ρ; the deterministic shape matrix alone supplies

no coverage. Shared calibration and cross-window dependence belong in that joint law or joint error region.

V. THE RETAINED CF4 QUOTIENT AND CONDITIONAL SHEAR BOUNDS

The retained calibration quotient fits a five-coordinate apparent directional-rate tensor using magnitude residuals. Its design already contains the optical conversion,

$$y_i = \mathrm{DMzp}_i - \mu_{\mathrm{ref},i}, \qquad B_{i\alpha} = -\kappa(n_i^{\mathsf{T}} T_\alpha n_i), \qquad \widehat{b} = Ky. \tag{23}$$

The coefficient $\widehat{b}$ is dimensionless, while y has magnitude units and K inverse magnitude units. The basis is

$$T_0 = \frac{\mathrm{diag}(1,-1,0)}{\sqrt{2}}, \quad T_1 = \frac{\mathrm{diag}(1,1,-2)}{\sqrt{6}}, \quad T_{xy}, T_{xz}, T_{yz} = \frac{e_i e_j^{\mathsf{T}} + e_j e_i^{\mathsf{T}}}{\sqrt{2}}. \tag{24}$$

Thus $\|\widehat{S}_{\mathrm{app}}\|_F = |\widehat{b}|$. The source uses J2000 Cartesian directions and a fixed reference cosmology. Its 35,970 selected rows include 944 mixed-support rows saturated by the nuisance quotient; the corrected exact factor has 945 zero-leverage rows. The saved target numerical rank is five, with singular values between 186.248352167 and 279.226623496 in its working metric [4]. This establishes survival of five apparent tensor directions in that finite design.

The retained matrix is

$$\widehat{S}_{\mathrm{app}} = \begin{pmatrix} -0.01737217048 & 0.00642017243 & 0.01454248664 \\ 0.00642017243 & 0.00395564447 & 0.00563999318 \\ 0.01454248664 & 0.00563999318 & 0.01341652601 \end{pmatrix}, \tag{25}$$

with

$$t := \|\widehat{S}_{\mathrm{app}}\|_F = 0.03265680189321492,$$
$$\lambda(\widehat{S}_{\mathrm{app}}) = \tag{26}$$
$$(-0.023727165269042373,\ 0.001327948995864026,\ 0.022399216273178365).$$

The directional RMS is $\sqrt{2/15}\, t = 0.011924578035260817$. The full precision matrix and coordinates, rather than the rounded display (25), generate the numerical table and figures. The magnitude quadrupole Q_μ is a different tensor: at leading order $Q_\mu = -\kappa S$, with

TABLE I. Retained apparent tensor in the dimensionless J2000 source basis (24). Singular values use the retained working metric. All central values are generated from full precision retained inputs.

Quantity	Value
b_0	-0.01508104258
b_1	-0.0164318214193
b_2	0.00907949492483
b_3	0.0205661818356
b_4	0.00797615484124
$lambda_1$	-0.023727165269
$lambda_2$	0.00132794899586
$lambda_3$	0.0223992162732
Frobenius norm	0.0326568018932
Directional RMS	0.0119245780353
Numerical rank	5
Minimum singular value	186.248352167
Maximum singular value	279.226623496

TABLE II. Conditional frontier for assumed r_Q and assumed $r_H/H_* = 0.1$. Rate columns mean $\|\sigma\|_F/(\sqrt{6}H_*)$. The zero-radius row is an exact-response limiting scenario, without a calibrated physical error estimate.

Assumed r_Q	Σ_-	Σ_+	Rate/$H_* : -$	Rate/$H_* : +$
0	0.013332083545	0.013332083545	0.01199887519	0.014665291899
0.005	0.011290842093	0.015373324997	0.010161757883	0.016910657497
0.01	0.0092496006403	0.01741456645	0.0083246405763	0.019156023095
0.02	0.0051671177356	0.021497049354	0.0046504059621	0.02364675429
0.032656801893	0	0.02666416709	0	0.029330583799
0.04	0	0.029662015163	0	0.03262821668

$$\|Q_\mu + \kappa S\|_F \le \sqrt{15/2}\,\frac{\kappa\epsilon^2}{2(1-\epsilon)}. \tag{27}$$

This follows by projecting (11) and using the quadrupole norm identity in Proposition 1. The conversion in (23) is already present in the CF4 producer. Dividing (26) by κ again would change its units convention and its physical scale.

Proposition 3 (shear-radius frontier). Suppose a response qualification on one common event, congruence and frame establishes

$$S = \sigma/H, \qquad \|S - \widehat{S}_{\rm app}\|_F \le r_Q, \qquad H \in [H_-, H_+], \quad 0 < H_- \le H_+. \tag{28}$$

The radius must cover the applicable statistical, calibration, sampling, finite-redshift, peculiar-motion and frame-response errors. Then the dimensionless shear amplitude and dimensional shear rate satisfy

$$\frac{(t - r_Q)_+}{\sqrt{6}} \le \Sigma := \frac{\|\sigma\|_F}{\sqrt{6}H} \le \frac{t + r_Q}{\sqrt{6}}, \tag{29}$$

$$\frac{H_-(t - r_Q)_+}{\sqrt{6}} \le \frac{\|\sigma\|_F}{\sqrt{6}} \le \frac{H_+(t + r_Q)}{\sqrt{6}}. \tag{30}$$

Proof. The triangle and reverse triangle inequalities give $(t - r_Q)_+ \le \|S\|_F \le t + r_Q$. Divide by $\sqrt{6}$ for (29) and multiply by a positive $H \in [H_-, H_+]$ for (30). Over the unrestricted STF ball, the two norm endpoints are attained along $\widehat{S}_{\rm app}$, or at zero if $r_Q \ge t$. Independent endpoint choices for H attain the dimensional endpoints over the corresponding product set. Additional physical or correlated-error restrictions can narrow these intervals. □

The lower endpoint is strictly positive exactly when $r_Q < t$. The present retained products do not estimate this qualified radius. Figure 1 therefore shows a requirement on the error budget, with r_Q/t and r_H/H_* as assumed sensitivity coordinates. In (30), write $H_\pm = H_* \pm r_H$, $0 \le r_H < H_*$, and divide the plotted dimensional shear rate by H_*. No numerical Hubble rate is needed.

If the measured tensor were instead a dimensional rate $\widehat{Q}_H$, with $\|\sigma - \widehat{Q}_H\|_F \le r_{Q_H}$, the normalized bounds would be

$$\frac{(\|\widehat{Q}_H\|_F - r_{Q_H})_+}{\sqrt{6}H_+} \le \Sigma \le \frac{\|\widehat{Q}_H\|_F + r_{Q_H}}{\sqrt{6}H_-}. \tag{31}$$

Equation (31) belongs to a different input convention. Applying its Hubble division to (26) would count the normalization twice. Ordered eigenvalues under (28) obey

$$|\lambda_i(S) - \lambda_i(\widehat{S}_{\rm app})| \le r_Q. \tag{32}$$

This follows from the min–max characterization and $\|S - \widehat{S}_{\rm app}\|_{\rm op} \le r_Q$. For example, the largest principal value is guaranteed positive by this bound only if $r_Q < 0.022399216273178365$. It refers

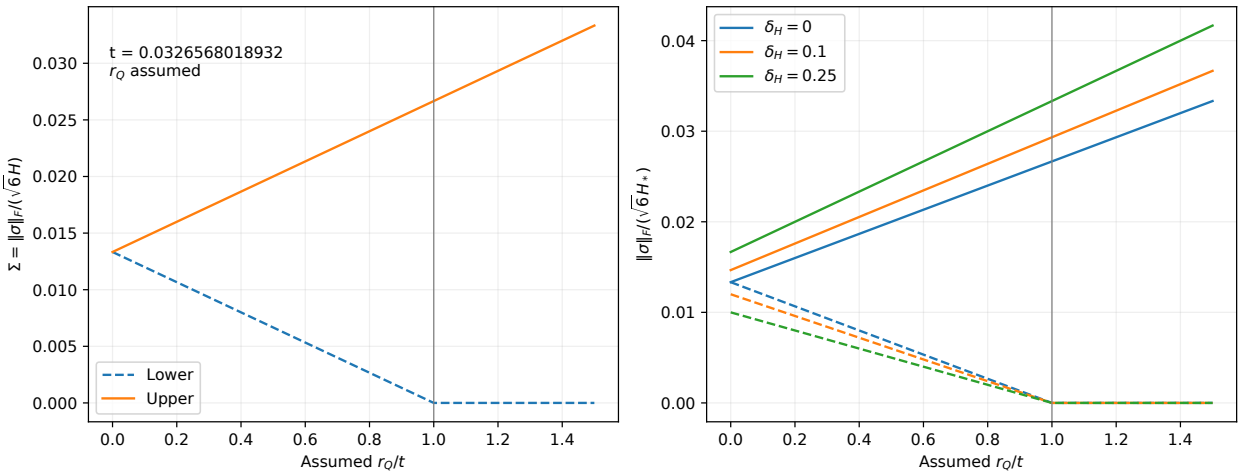


FIG. 1. Conditional shear bounds from the retained apparent CF4 tensor. The horizontal coordinate is an assumed qualified dimensionless response-error radius divided by the apparent norm. The left panel gives the dimensionless shear-amplitude interval. The right panel converts it to a dimensional shear rate, divided by a symbolic reference Hubble rate, for three assumed relative Hubble intervals. A positive lower bound requires r_Q below the apparent norm. The figure assumes the same-event, same-congruence response in equation (28); neither error radius nor a physical shear detection is estimated here.

to expansion relative to the local mean. Eigenvector directions require an additional gap-sensitive perturbation analysis.

Figure 2 retains more information than the ball in (29): it shows a two-coordinate projection of (21), using the saved working shape matrix with $\rho = 1$, $G = I$, and selected assumed η. The curves are deterministic images of those chosen sets. They have no assigned confidence level.

VI. TENSOR CARRIERS AND UNRESOLVED PHYSICAL SECTORS

Let Z retain the tensor state, support and response parameters, rather than only its norm. A suitable optical carrier and its deterministic image are

$$x_\phi(Z) = (d, S, \omega/H, H, \text{other physical sectors}), \qquad x_\phi[\Theta] = \{x_\phi(Z) : Z \in \Theta\}. \tag{33}$$

For a finite linear model Θ includes the kernel in (17); for the complete field, d, S are determined but H and vorticity remain unresolved. The local slope contains no antisymmetric velocity gradient, and the angular normalization removes the absolute expansion scale. Curvature, a matter-frame energy split and a temporal history are also absent. The signed comparator $x_C = \Sigma^2 - W^2 + \Omega_{\text{tilt}} + \Omega_{k,\text{aniso}}$ consequently remains incomplete when only the shear sector is bounded. Here the

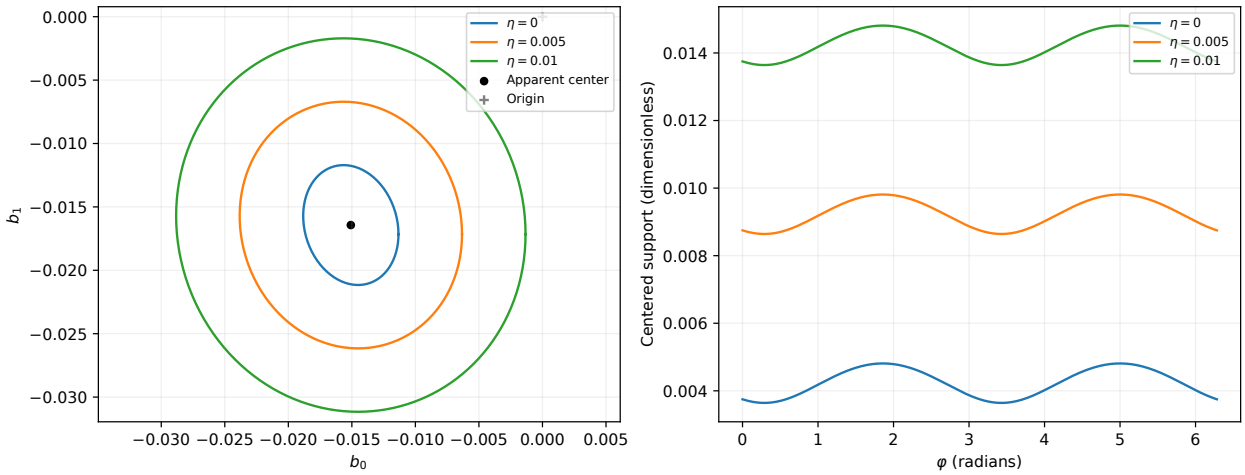


FIG. 2. A deterministic projection of the tensor error region. The retained working matrix supplies an illustrative ellipsoidal shape, with unit assumed ellipsoid radius and three assumed isotropic response radii. Curves show the projection onto the first two Frobenius-orthonormal source coordinates, and the corresponding centered directional support. The product error-set assumption gives the support exactly. The unit radius has no assigned confidence level, and the additional response radii are sensitivity choices.

comparator quantity W^2 is the dimensionless vorticity contribution; it is distinct from the working whitening matrix W in Section 3. Unmeasured terms are not assigned zero.

For a declared positive tolerance q_*, one may form a policy score, for example

$$Q_\phi(Z) = \|S\|_F / q_*, \qquad q_* > 0. \tag{34}$$

Its interval image follows from the tensor region. The denominator is a policy input, not a fitted significance. At zero tolerance, use an exact set constraint instead of defining a ratio. The remaining diagnostics require different inputs:

$$F_\nu(B) = \frac{\nu(B \cap D)}{\nu(D)}, \quad 0 < \nu(D) < \infty; \qquad \Pi_P(a) = P\{Q_\phi(Z) > a\}; \qquad G_F = \mathcal{G}(F_1, \ldots, F_k). \tag{35}$$

Occupancy F requires a support measure and a specified admissible set or sign-clean ceiling. Exceedance Π requires a named joint law. A depth functional G_F requires registered windows, their measures and shared dependence. Here x_ϕ has an apparent measured tensor component and conditional physical image; Q is specified only after q_* is chosen; F, Π, G_F are undefined because the corresponding measure, law and depth functional have not been supplied. The saved working

matrix does not supply these missing objects. A model-dependent probability pushforward belongs to HTT; the deterministic region can be reported as a MIO diagnostic.

VII. DISCUSSION

There are two useful inverses. A complete positive local magnitude field determines its dimensionless acceleration and shear after normalized exponentiation. A finite observation first passes through sampling and nuisance removal, leaving an identifiable subspace, a kernel and a residual-compatibility condition. Their error bounds answer different questions, so the full-field identity cannot be substituted into an incomplete map without controlling the missing field.

The retained CF4 quotient makes the error target concrete. Its apparent tensor norm is 0.0326568 in the producer's dimensionless convention. To obtain a positive lower shear bound through the stated response identity, a qualified common-event radius must lie below this scale. A supplied response region can be propagated with (18) or (21), retaining anisotropy and dependence before taking the simpler norm bounds. An absolute Hubble interval then converts a dimensionless shear range to a dimensional rate.

This construction isolates what an additional physical input must do: constrain the response error or a currently unresolved sector. It also permits a complete conditional inverse result while those stronger inputs remain unavailable. The attainable sets and their support functions are the methods result; the apparent tensor is its retained finite-data illustration.

Appendix A: Joint optical residuals with a separate direct-dipole constraint

1. The seven-component response and its physical domain

The finite inverse can retain the complete signed octupole when testing a common local optical response. We use the same outward direction, signature $(-+++)$, and orthonormal observer rest frame as in the main text. With $H > 0$, $d = -A/(cH)$, $S = \sigma/H$, and $\kappa = 5/\ln 10$, the local field is $-\kappa \ln[1 + d \cdot n + S : nn]$. The quantities H and σ have inverse-time units; d, S are dimensionless. A physical application assumes one smooth emitter congruence and a regular local distance expansion. Let $\sup_n |d \cdot n + S : nn| \le \epsilon < 1$ throughout the proposed parameter region. This is a condition on the region, not merely on its central estimate.

In Frobenius-orthonormal rank-three symmetric trace-free coordinates, the magnitude octupole has seven signed components. The retained local expansion gives [3]

$$O = \kappa L_S d + R_3, \qquad L_S d = \mathrm{STF}(d \otimes S), \qquad \|R_3\|_F \le E_3(\epsilon) = \sqrt{\frac{35}{2}} \frac{|\kappa|\epsilon^3}{3(1-\epsilon)}. \tag{A1}$$

The response has the Gram matrix

$$M_S = L_S^T L_S = \frac{S:S}{3} I + \frac{2}{5} S^2. \tag{A2}$$

For nonzero S, this three-dimensional matrix is positive definite, while the seven-dimensional octupole still has four directions perpendicular to the response image. The measured apparent CF4 tensor retained in the main text has Frobenius norm 0.03265680189321492. Inserting that same dimensionless tensor gives minimum and maximum singular values of L_S equal to 0.018873110043440326 and 0.024097307981999622. These quantify the conditional linear inverse. The tensor is already dimensionless and is not divided by κ again; its physical interpretation as σ/H still requires the stated response qualification.

Suppose $\|S - \widehat{S}\|_F \le e_S$ and $\|d\| \le D_{\max}$ hold on the same deterministic region or joint event. The STF identity implies $\|L_{S-\widehat{S}}\|_{\mathrm{op}} \le \sqrt{3/5}\, e_S$. A single outer radius for the seven-component optical residual is therefore

$$B = E_3(\epsilon) + e_O + B_z + B_{\mathrm{sel}} + |\kappa| \sqrt{3/5}\, e_S D_{\max}. \tag{A3}$$

Here e_O bounds the remaining observational octupole error, B_z the finite-distance contribution, and B_{sel} the effect of selection, calibration and the specified fitting operator. Each contribution is included once according to its definition. The angular logarithmic remainder does not supply the finite-distance or selection bounds. A sufficient uniform domain condition is $D_{\max} + \|\widehat{S}\|_{\mathrm{op}} + e_S \le \epsilon < 1$. More restrictive joint regions can also establish the required bound. A nonzero uncertain-shear term requires a finite bound on $\|d\|$; omitting that bound cannot silently set the term to zero.

A separately supplied direct-dipole region constrains the argument of the optical map. For a spherical direct region and an optional physical amplitude bound, write

$$\mathcal{D} = \{d : \|d - d_{\mathrm{obs}}\| \le r_d\} \cap \{d : \|d\| \le D_{\max}\}, \qquad r_* = \min_{d \in \mathcal{D}} \|\widehat{O} - \kappa L_{\widehat{S}} d\|_F. \tag{A4}$$

The second ball may be omitted in the purely mathematical fixed-response problem. Physical use still requires the uniform angular domain and any amplitude bound used to construct B.

Compatibility with the outer residual model means $\mathcal{D} \neq \varnothing$ and $r_* \leq B$. The direct-dipole radius r_d is profiled through $\mathcal{D}$; it is not an additional octupole-error term in B. A supplied correlated error body may be more restrictive than these separate outer bounds.

For $\widehat{S} \neq 0$, set $\widehat{d} = \kappa^{-1} M_{\widehat{S}}^{-1} L_{\widehat{S}}^T \widehat{O}$ and $\widehat{O}_\perp = \widehat{O} - \kappa L_{\widehat{S}} \widehat{d}$. At every admissible d,

$$\|\widehat{O} - \kappa L_{\widehat{S}} d\|_F^2 = \|\widehat{O}_\perp\|_F^2 + \kappa^2 (\widehat{d} - d)^T M_{\widehat{S}} (\widehat{d} - d). \tag{A5}$$

Both terms spend the same squared residual budget. Two orthogonal contributions of $0.8B$ each give $\sqrt{1.28}B > B$, although each passes its separate threshold. The saved unit-budget example has certified residual endpoints 1.1313708498984756 and 1.1313708498984758 and is incompatible. At $S = 0$, the response is zero and the appropriate test is directly $\|\widehat{O}\|_F \leq B$, together with nonemptiness of $\mathcal{D}$. No inverse is used. At $B = 0$, retain exact membership instead of a normalized score. Arbitrarily small nonzero S remains a distinct inverse problem; an absolute numerical threshold cannot replace it by $S = 0$.

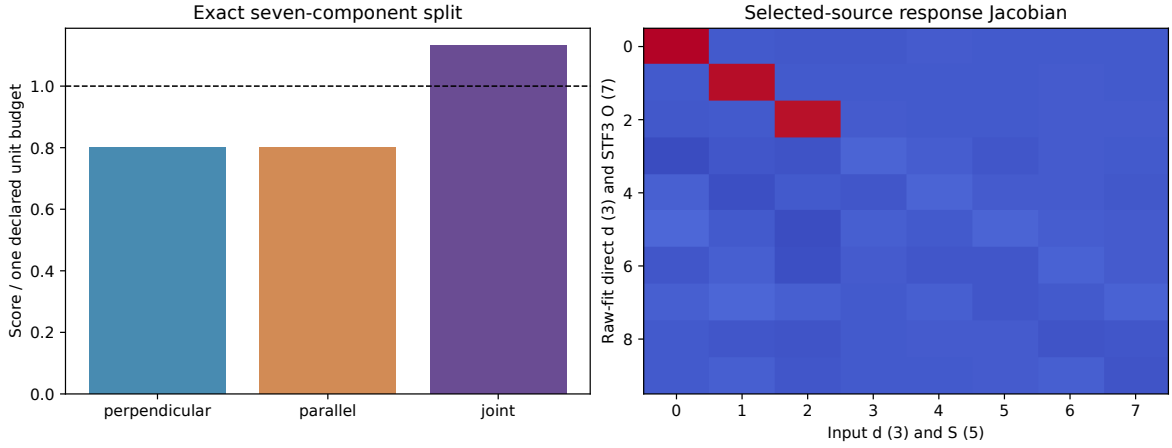


FIG. 3. A shared optical residual budget and a finite selected-source response. Left: the two orthogonal residual components each have size 0.8 of one declared unit budget; their combined norm exceeds that budget. Right: the saved 10-by-8 local Jacobian maps three dimensionless dipole and five dimensionless shear coordinates to three fitted direct-dipole coordinates and seven magnitude-octupole coordinates. Both panels are deterministic mathematical controls. The sampled response is not identified with a physical catalogue dipole.

2. Sampling, covariance and finite distance

Before assigning a physical meaning to fitted harmonics, retain the map from source magnitudes. Let $T_a^{(2)}$ and $T_b^{(3)}$ be Frobenius-orthonormal STF bases. For directions n_i and positive weights w_i, use the full-rank 16-column design

$$X_i = (1, n_i^T, T_a^{(2)} : n_i n_i, T_b^{(3)} : n_i n_i n_i), \qquad K = J(X^T W X)^{-1} X^T W, \quad W = \mathrm{diag}(w_i). \tag{A6}$$

The selector J multiplies the three fitted magnitude-dipole coordinates by $-1/\kappa$ and retains all seven fitted octupole coordinates. Thus Ky is a ten-coordinate fitted summary; its first three entries need not equal the physical d. For the local source model $y_i = -\kappa \ln(1 + q_i)$, $q_i = d \cdot n_i + S : n_i n_i$, its derivative is

$$\frac{\partial(Ky)}{\partial(d, S)} = K \,\mathrm{diag}\left(\frac{-\kappa}{1+q_i}\right) [\, n_i^T,\ T_a^{(2)} : n_i n_i \,]. \tag{A7}$$

The fixed illustration uses 96 deterministic directions, weights $1 + 0.4 n_z$, and an input vector $(0.012, -0.009, 0.006)$, with the retained apparent tensor inserted as a mathematical coefficient. The rank of its sampled design is 16. The direction index is $i = 0, \ldots, 95$, with $z_i = 1 - 2(i + 1/2)/96$, $\phi_i = i\pi(3 - \sqrt{5})$, and $n_i = (\sqrt{1 - z_i^2}\cos\phi_i, \sqrt{1 - z_i^2}\sin\phi_i, z_i)$. These directions and coefficients are an illustration, not a CF4 data selection or a measured acceleration.

The signed coordinate convention in this illustration is fixed by lexicographic Cartesian unit monomials, STF projection, and successive Frobenius orthogonalization with positive normalization. The full rank-two and rank-three basis arrays accompany the numerical supplement. This rank-two coordinate ordering differs from the five-coordinate display in the main text; the common object is the Cartesian tensor S, not an untransformed coordinate list.

For a supplied full source covariance C_y, the fitted covariance is $C_{10} = K C_y K^T$. At fixed $\widehat{S}$, its contribution to the residual covariance is

$$C_r = A C_{10} A^T, \qquad A = [-\kappa L_{\widehat{S}} \quad I_7]. \tag{A8}$$

The off-diagonal direct-dipole–octupole blocks contribute to this expression. The illustration stipulates $C_y = 10^{-8}(I + 0.2uu^T)\,\mathrm{mag}^2$, where the 96 components of u are equally spaced from -1 to 1. Its direct-dipole–octupole cross-block Frobenius norm is $1.7820478375045505 \times 10^{-12}$ in

TABLE III. Complete signed response of the declared 96-direction mathematical illustration. The seven octupole entries use the explicit STF basis in the numerical supplement; neither the fitted direct vector nor this octupole is a measured CF4 physical quantity.

Fitted coordinate	Signed value	Unit
d_x	+0.012074086313833	dimensionless
d_y	-0.00903120168213684	dimensionless
d_z	+0.00591810638421224	dimensionless
O_1	-0.000548756550802814	mag
O_2	+0.00040268605514402	mag
O_3	+0.000221664713143214	mag
O_4	-0.000393290392446293	mag
O_5	-4.62789847900094e-06	mag
O_6	+9.87390973288633e-06	mag
O_7	-0.000171094484684956	mag

dimensionless-dipole times magnitude units, and $\operatorname{tr} C_r = 1.317508814238013 \times 10^{-8}\,\mathrm{mag}^2$. The complete 10-by-10 covariance, 7-by-7 residual covariance and 10-by-8 response derivative are retained, with their coordinate order and units. This is a source-covariance illustration at fixed S, not a supplied empirical CF4 covariance, not an uncertain-S covariance calculation, and not a confidence region. A confidence threshold would additionally require a sampling law and its calibration. No unknown empirical cross-block is filled with zero.

Finite distance requires a separate contribution even when the angular expansion is controlled. If

$$d_L = \frac{cz}{H(1+q)}[1 + za_1(n) + O(z^2)], \tag{A9}$$

then the magnitude correction is $\kappa z a_1(n) + O(z^2)$, generally first order in redshift. The saved source-level control uses $a_1(n) = T_1^{(3)} : nnn$ and passes $\kappa \ln[1 + za_1(n)]$ through the same fitted octupole operator. At $z = (0.02, 0.01, 0.005)$, its octupole norms are $(0.04342971604250298, 0.021714693414132947, 0.010857342173346137)$ mag. The successive ratios 2.0000151608973153 and 2.0000008351436778 illustrate the leading linear order. These values do not determine a physical B_z for a survey; that bound needs a controlled finite-distance coefficient and the actual estimator and depth weights.

Observer motion also changes the target. The same-congruence coefficient $d = -A/(cH)$ is not interchangeable with a peculiar-velocity dipole. For the dust moving-observer construction,

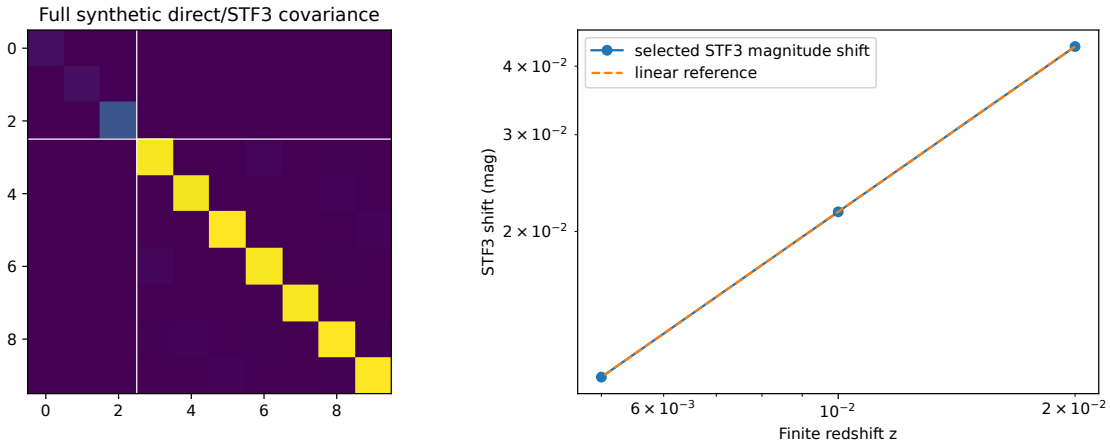


FIG. 4. Declared source covariance and a finite-redshift response control. Left: all entries of the synthetic ten-coordinate covariance, ordered as three fitted dimensionless direct-dipole coordinates followed by seven magnitude-octupole coordinates. Entry units follow the corresponding pair of coordinates; the white lines mark the shared-source cross-blocks. The full numerical matrix accompanies the text. Right: the same selected octupole operator applied to a declared relative-distance perturbation, compared with a linear-redshift reference. Neither panel supplies an empirical CF4 joint law or a physical finite-distance error radius.

the first-order relation is $d_{\mathrm{boost}} = -2(I+S)\beta + O(\|\beta\|^2)$, with $\beta = v/c$; its inverse requires invertibility of $I+S$ and control of the omitted order. At a fixed ray, the Doppler factor $\Gamma > 0$ gives $1+\widetilde{z} = (1+z)/\Gamma$, $\widetilde{d}_L = d_L/\Gamma$, and a shifted vertex $\widetilde{z}_o = \Gamma^{-1} - 1$. The corrected distance $d_L^* = d_L/(1+z)^2$ obeys $\widetilde{d}_L^* = \Gamma d_L^*$, so the vertex derivative transforms as $d\widetilde{z}/d\widetilde{d}_L^* = \Gamma^{-2} dz/dd_L^*$ [2]. The uncorrected-distance artefact $2\,\mathrm{STF}(\beta \otimes S)$ is distinct from the nonlinear magnitude octupole $\kappa L_S d$. Neither relation turns a raw catalogue harmonic into a measured acceleration or observer velocity.

3. Closed sets and numerical certificates

The numerical compatibility calculation keeps the two original dipole balls closed. Finite binary64 coordinates and radii are treated as exact rational inputs for squared-ball membership and emptiness. Zero-radius balls and external tangency are handled before optimization. A rounded tangent point is accepted only if it satisfies both original predicates. Failure to represent or verify a feasible witness returns numerical uncertainty, not physical exclusion or an enlarged radius.

An optimizer supplies a candidate, not a compatibility certificate. A verified feasible candidate supplies a primal residual bound. A lower bound follows from the seven-component support

inequality: for $\|u\| \le 1$,

$$r_* \ge u^T \widehat{O} - h_{\mathcal{D}}(K_S^T u), \qquad K_S = \kappa L_{\widehat{S}}, \tag{A10}$$

where the support function obeys

$$h_{\mathcal{D}}(v) \le \min\{v^T d_{\mathrm{obs}} + r_d\|v\|, D_{\max}\|v\|\}. \tag{A11}$$

If the amplitude ball is omitted, only the first expression is used. This support bound need not be sharp. The implementation uses candidate residual and observation directions and incorporates its declared response-roundoff envelopes. It exports a lower endpoint rounded downward and an upper endpoint rounded upward. The positive-budget score is the upward-rounded quotient of the exported upper endpoint and B; it is an upper bound, not a claim to have computed the exact minimizing score.

For a general nondegenerate case, the implemented decisions compare the upper endpoint before display rounding with $B + \delta_{\mathrm{loc}}$, or the lower endpoint with $B + \delta_{\mathrm{glob}}$. With binary64 machine epsilon $\varepsilon_{\mathrm{mach}}$, the cancellation allowance is $\delta_{\mathrm{loc}} = 64\varepsilon_{\mathrm{mach}} \max(\|\widehat{O}\|, \|K_S d_w\|)$ only when both terms are nonzero and the witness residual is at most half the larger norm; otherwise it is zero. The conservative rejection allowance is $\delta_{\mathrm{glob}} = 64\varepsilon_{\mathrm{mach}} \max(\|\widehat{O}\|, \|K_S\|_F D_{\mathrm{cap}})$, where D_{cap} is the upward-enclosed triangle cap $\|d_{\mathrm{obs}}\| + r_d$, restricted by $D_{\max}$ when supplied. These are numerical response allowances, separate from the physical radii. A positive cancellation allowance is explicitly recorded, and an accepted case must not be advertised as a strict zero-allowance residual certificate unless its exported upper bound also lies below B. A straddling interval remains numerical uncertainty.

For a singleton set or zero response, the finite-response residual square is compared with B^2 using exact rational arithmetic, including equality. A nonzero-response exclusion too close to the declared global response allowance is still returned as numerical uncertainty. Underflow of a displayed square does not set a nonzero residual to zero. The complete certificates retain the witness, endpoint bounds, response envelope, cancellation allowance and display status.

The near-tangent examples make the closed-set convention concrete. Set both radii to $1/8$. A center $(1/4, 0, 0)$ has the representable unique intersection $(1/8, 0, 0)$. For the adjacent binary64 center below $1/4$, a feasible saved witness is $(0.12499999999999999, 0, 0)$; with $S = \mathrm{diag}(0.001, -0.001, 0)$, zero observed octupole and $B = 0.001$, its exported upper residual

TABLE IV. Saved finite-response controls. Endpoints bound the profiled residual or the exact fixed objective under the stated numerical response conventions; they are not uncertainty intervals for physical cosmological parameters. Empty-set decisions precede residual comparison. All radii are stipulated. Full witnesses and all diagnostic fields accompany the numerical supplement.

Case	Lower	Upper	B	Result
Shared unit budget	1.1313708498984756	1.1313708498984758	1.0	Incompatible
Zero response, zero budget	0.0	0.0	0.0	Compatible
Narrow feasible ball	0.0	5.176664520037313e-19	5.606717652542515e-10	Compatible
Excluded physical boundary	2.0184131778458593e-15	2.0184235050681796e-15	1.0092091825422658e-15	Incompatible
Tangent singleton	0.0002803358826268454	0.00028033588262684546	0.001	Compatible
One-spacing interior overlap	0.00028033588262683885	0.00028033588262685186	0.001	Compatible
Zero response, fixed sqrt(3) norm	1.7320508075688772	1.7320508075688774	2.0	Compatible
Empty direct singleton	—	—	—	Empty
Adjacent exterior empty set	—	—	—	Empty

bound is $0.00028033588262685186 < B$. The adjacent center above $1/4$ gives an exactly empty intersection. A direct singleton at $(1/8, 10^{-60}, 0)$ lies outside the physical ball of radius $1/8$: the second component gives a strictly positive exact squared excess, even when a rounded norm hides it. The decimal 10^{-60} here denotes its supplied binary64 value, whose square is near 10^{-120}, not exactly that decimal rational. Zero and nonzero shear controls also preserve a residual of 10^{-200} against a zero budget. These examples test witness availability and range handling; they do not prove that every feasible real intersection has a representable witness or that every numerical optimization is decisive.

The physical output remains a conditional response-compatible set. The apparent tensor, source-covariance illustration and numerical controls do not supply an empirical joint direct-dipole–octupole law, the missing finite-distance and selection bounds, or a common physical source/observer transfer. Vorticity, curvature, energy, absolute expansion and history remain unresolved as described in the main text. Full signed coordinates and shared dependence must be retained before any later scalar summary or physical attribution.

[1] A. Heinesen, "Multipole decomposition of the general luminosity distance 'Hubble law'—a new framework for observational cosmology," JCAP 05, 008 (2021), DOI 10.1088/1475-7516/2021/05/008, arXiv:2010.06534v2. https://arxiv.org/abs/2010.06534v2

[2] R. Maartens, J. Santiago, C. Clarkson, B. Kalbouneh and C. Marinoni, "Covariant cosmography: the observer-dependence of the Hubble parameter," JCAP 09, 070 (2024), DOI 10.1088/1475-

7516/2024/09/070, arXiv:2312.09875v3. https://arxiv.org/abs/2312.09875v3

[3] "Sharp quadrupole–octupole chirality and the nonlinear local distance response," retained companion methods manuscript, R32-01-QO-METHODS (2026), equations (20)–(24), (36), (38)–(39); repository source products/R32-01-QO-METHODS/science/MANUSCRIPT.md. This is a repository manuscript, not an asserted journal publication.

[4] Source-bound CF4 calibration-quotient sufficient statistics, EVIDENCE_MANIFEST.json and evidence/research/latest_results.json, retained release 25 October 2026, especially units_correction and cf4_calibration_quotient. Original numerical producer R28 at commit 71b272db893a92ec336c28164bc7efccf7c340b2; corrected interpretation bound to model SHA-256 843b19e464817d14455df68e9043a93cc71cc7449ab339052c3ddfe1c7298c19. These are retained numerical research products, not an independently calibrated physical error law.